

Stories from a City in Nature

1. The monkey and the hairless rambutan
2. Helping ~~grandma~~ wild boar, sambar deer, monkeys, pangolin cross the road using AI
3. Mee Siam - Pigeon's favourite food
4. The bird that needed a thigh massage



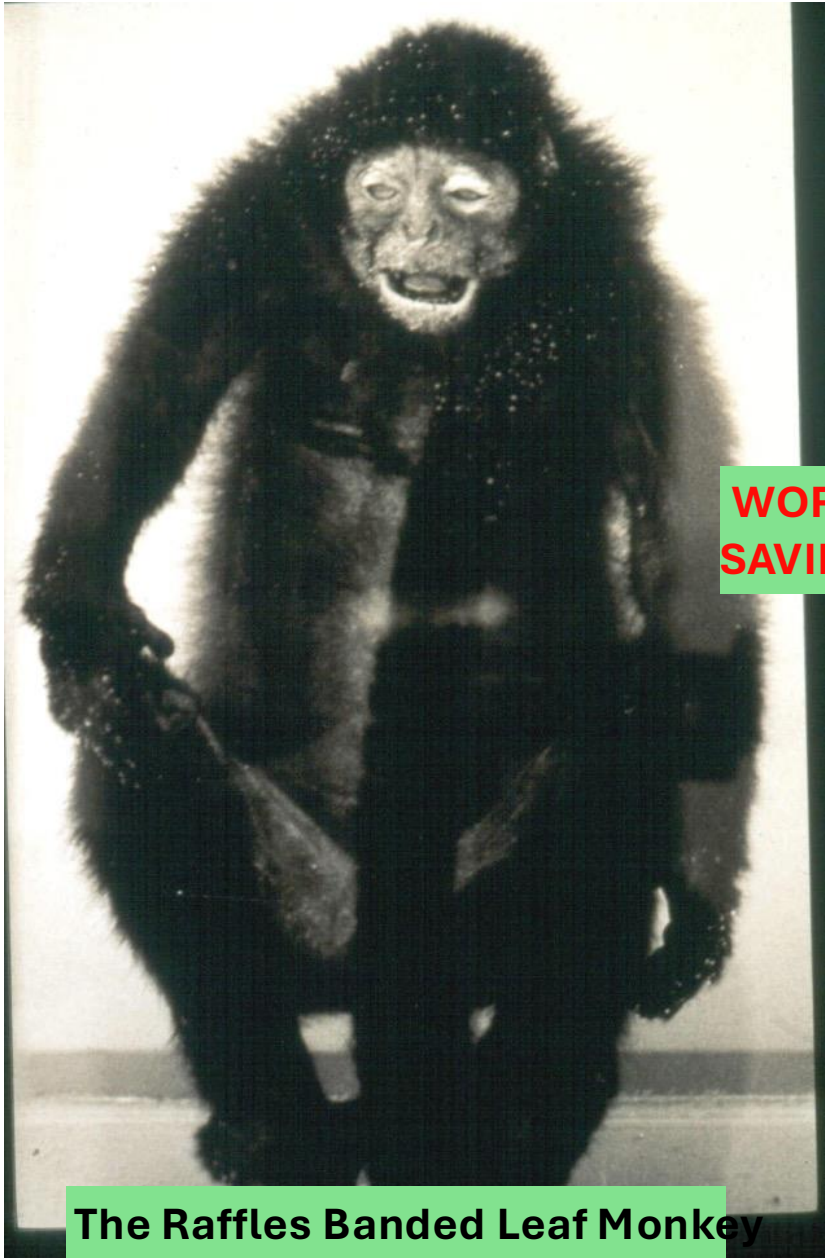
Adrian Loo

*Deputy Director, Centre for Nature-based Climate Solutions,
National University of Singapore*

1. The monkey and the hairless rambutan



THE LIVING DEAD



WORTH
SAVING ?

Critically endangered Raffles' banded langur found as roadkill in Upper Thomson Road



One monkey had managed to cross the road before the second one was run over. PHOTO: ZEN MA/FACEBOOK



Jolene Ang

PUBLISHED 27 AUG 2021, 11:05 PM SGT



SINGAPORE - A critically endangered Raffles' banded langur was spotted as roadkill in Upper Thomson Road on Friday morning (Aug 27).

Primatologist Andie Ang, chair of the Raffles' banded langur working group, said that with the death of the animal, there are only 68 such monkeys left in Singapore.



Last mile problem

Urban Analysis Exercise – Use SWOT to interpret your observation to problem statement

Strengths

- there are RBLs in Singapore
- they are crossing the tree canopy over the road

Weaknesses

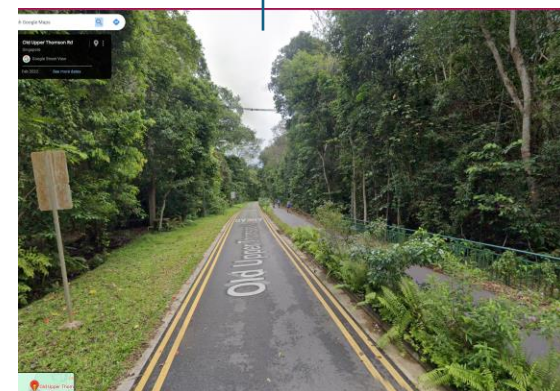
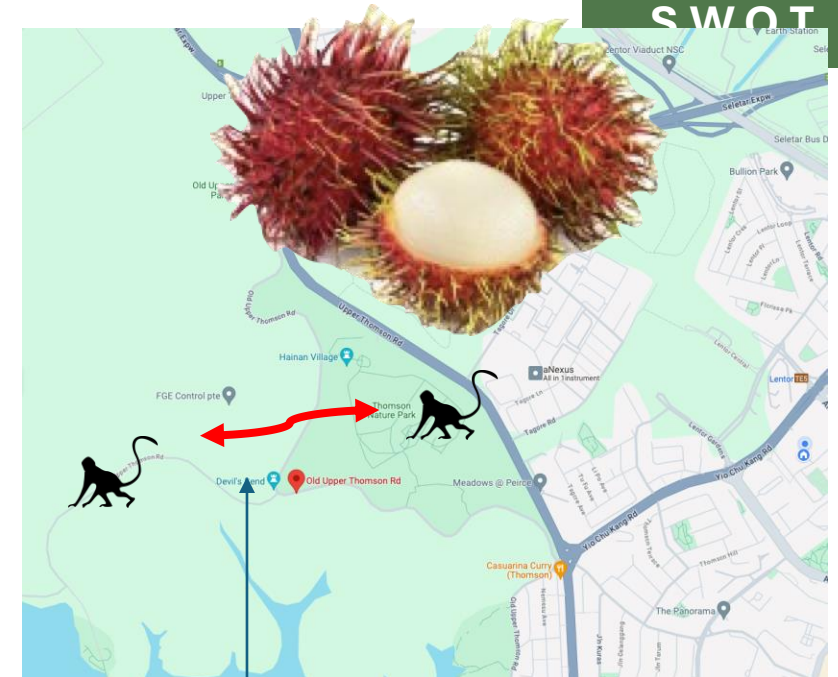
- the trees are sometimes pruned – no connection, RBLs cross road
- Cannot reach rambutans safely!

Opportunities

- leverage on technology
- study the current population numbers
- potential to build goodwill with stakeholders (NGOs)
- rescue the species from extinction

Threats

- lead to local extinction of RBLs
- reputation of Nparks, LTA





Raffles' Banded Langur crossing rope bridge at Thomson Nature Park

yahoo/news | Yahoo News Singapore

Highly endangered Raffles' Banded Langur population in Singapore to double in 15 years

New research shows the population of Raffles' Banded Langurs is projected to grow from 75 to over 150 in the next 15 years, and reach around 244 by 2071.



Ronald Goh · News and Lifestyle Producer

10 October 2023 · 4-min read

In this article:

Stamford Raffles

British statesman who founded Singapore



Highly endangered Raffles' Banded Langur population in Singapore to rise to 244 by 2071 (Photos: Dr Andie Ang)

SINGAPORE - The [critically endangered Raffles' Banded Langur](#), once thought to be on the brink of extinction in Singapore, is now projected to have a population of around 244

2. Helping wild boar, sambar deer, monkeys,
pangolin cross the road using AI

Minimising Human-Wildlife Conflict

Roadside Animal Detection System (RADS)

- Implemented in Florida and the Netherlands
- RADS alerts motorists when large animals, like panthers, are close to the road and advises drivers to slow down

Wild sambar deer put down due to severe injuries after 3-vehicle accident on BKE



Police shoot wild boar that was rampaging in Punggol



1 of 2 The wild boar, which was later euthanised due to its injuries, according to Acres. PHOTO: ACRES



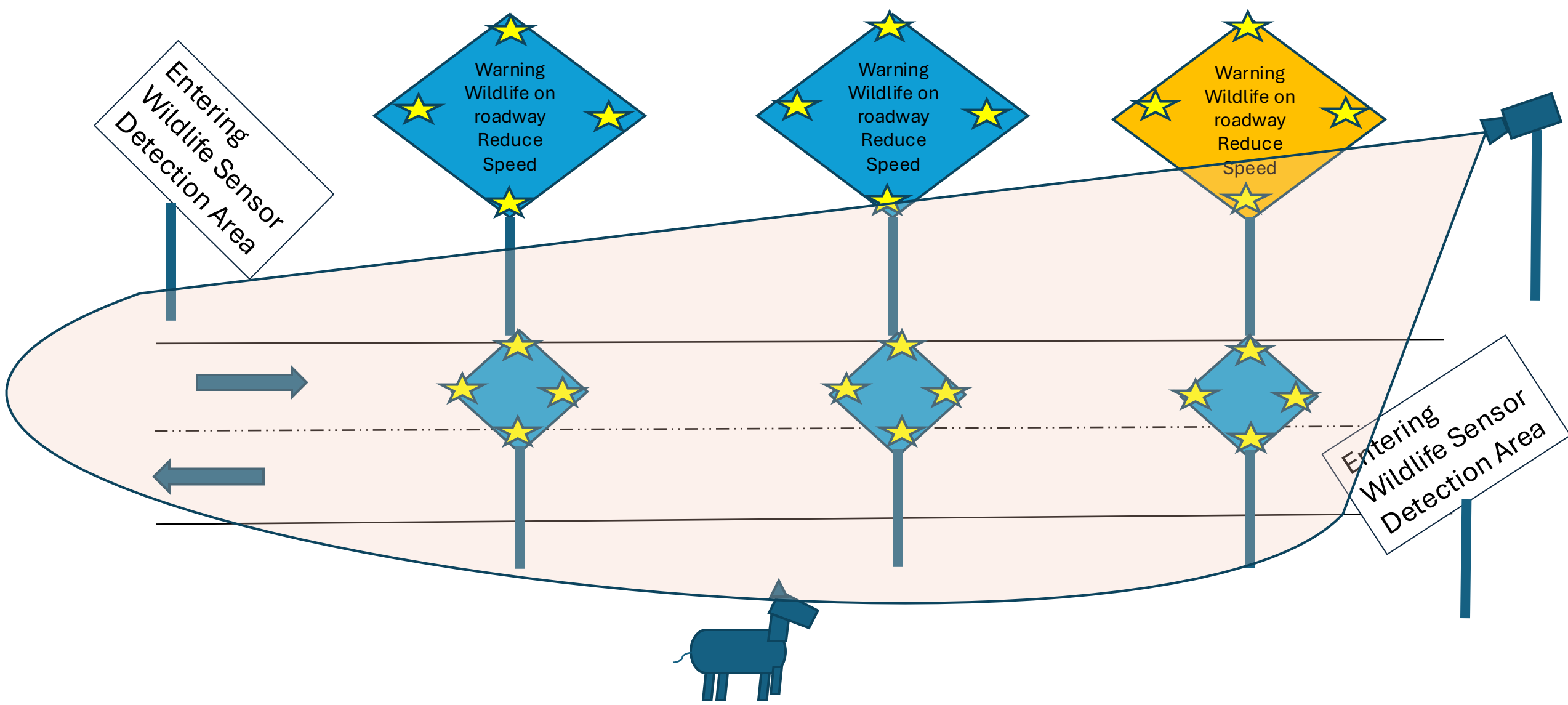
Roadside Animal Detection System (RADs)

Informational sign

Flashing sign

Flashing sign

Flashing sign



Roadway Animal Detection System (RADs)

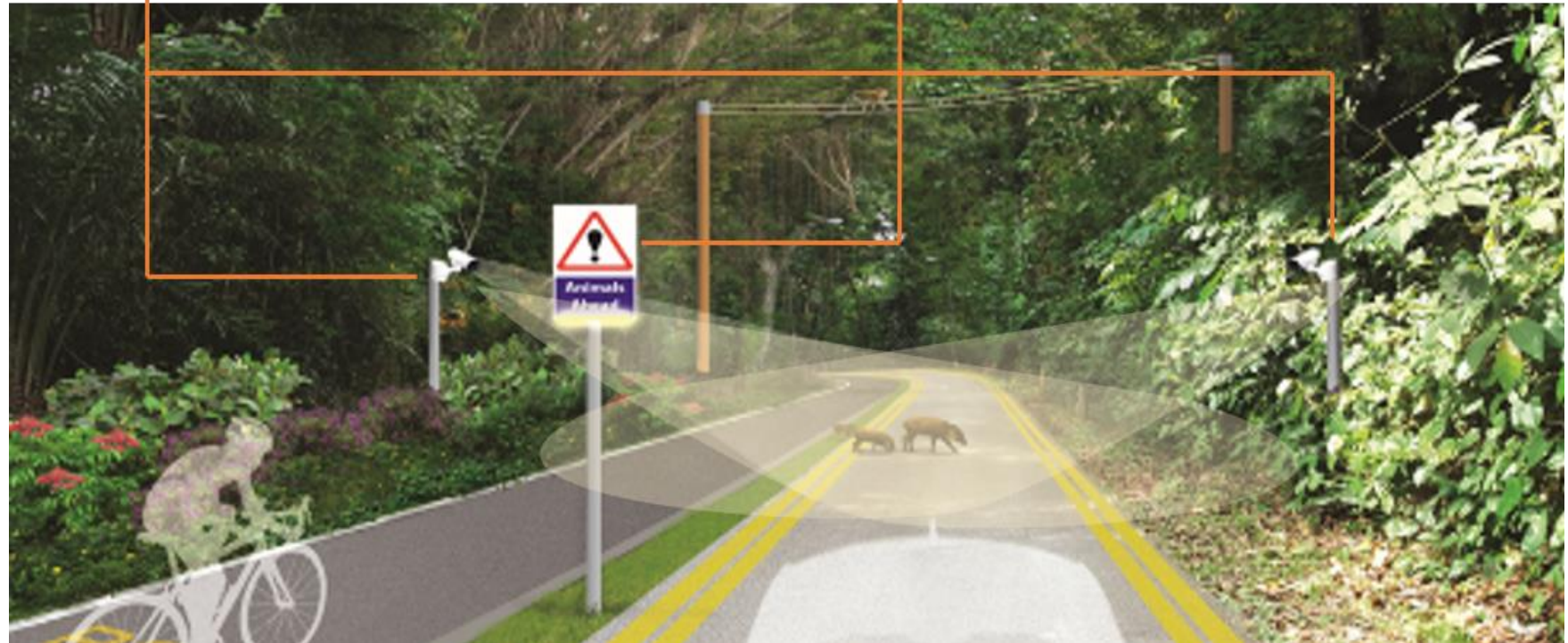
- Detection of animal movement triggers alert in the form of flashing lights on signboard to alert approaching motorists
- Video analytics applied on CCTV footage differentiates humans from animals

1

CCTV detects animals crossing the road using video analytics

2

Alerts motorist with flashing signs



3. Mee Siam - Pigeon's favourite food



Restricted human activities shift the foraging strategies of feral pigeons (*Columba livia*) and three other commensal bird species

Malcolm C.K. Soh  , Roanna Y.T. Pang, Breyll X.K. Ng, Benjamin P.Y.-H. Lee , Adrian H.B. Loo , Kenneth B.H. Er 

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<https://doi.org/10.1016/j.biocon.2020.108927>

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Abstract

Invasive species are a growing concern with increasing global connectivity. Feral pigeons (*Columba livia*) are widespread and invasive, thus their effective control is of keen international interest. The COVID-19 pandemic has offered an unprecedented

Range of measures used to control population of invasive rock pigeons in Singapore



Received: 11 August 2022 | Accepted: 22 March 2023

DOI: 10.1002/2688-8319.12234

RESEARCH ARTICLE

Ecological Solutions
and Evidence



Proximity to anthropogenic food sources determine roosting and nesting prevalence of feral pigeons (*Columba livia*) in a tropical city

Kai Ning Lim  | Malcolm C. K. Soh  | Duncan Y. W. Leong  | Adrian H. B. Loo  | Benjamin P. Y.-H. Lee  | Kenneth B. H. Er 

National Parks Board, Singapore Botanical Gardens, Singapore

Correspondence
Malcolm C. K. Soh
Email: malcolm_soh@nparks.gov.sg

Funding information
National Parks Board

Handling Editor: Juniper Simonis

Abstract

1. The feral pigeon (*Columba livia*) is a globally commensal bird that can cause disamenities such as soiling and is a potential vector of various diseases. Aside from limiting food as a management strategy, reducing the availability of roosting and nesting sites can help regulate feral pigeon populations. Despite their prevalence, current knowledge of their roosting and nesting preferences is lacking.
2. Feral pigeons commonly use railway viaduct expansion gaps in Singapore for roosting and nesting. These gaps provided an ideal experimental platform to examine feral pigeon roost and nest site selection while controlling for differing cavity sizes which can significantly affect their reproductive success and site se-



An increase in wild boar population has lead to more human-wild boar conflict, according to NParks.

2. February 2019, Lower Seletar Reservoir

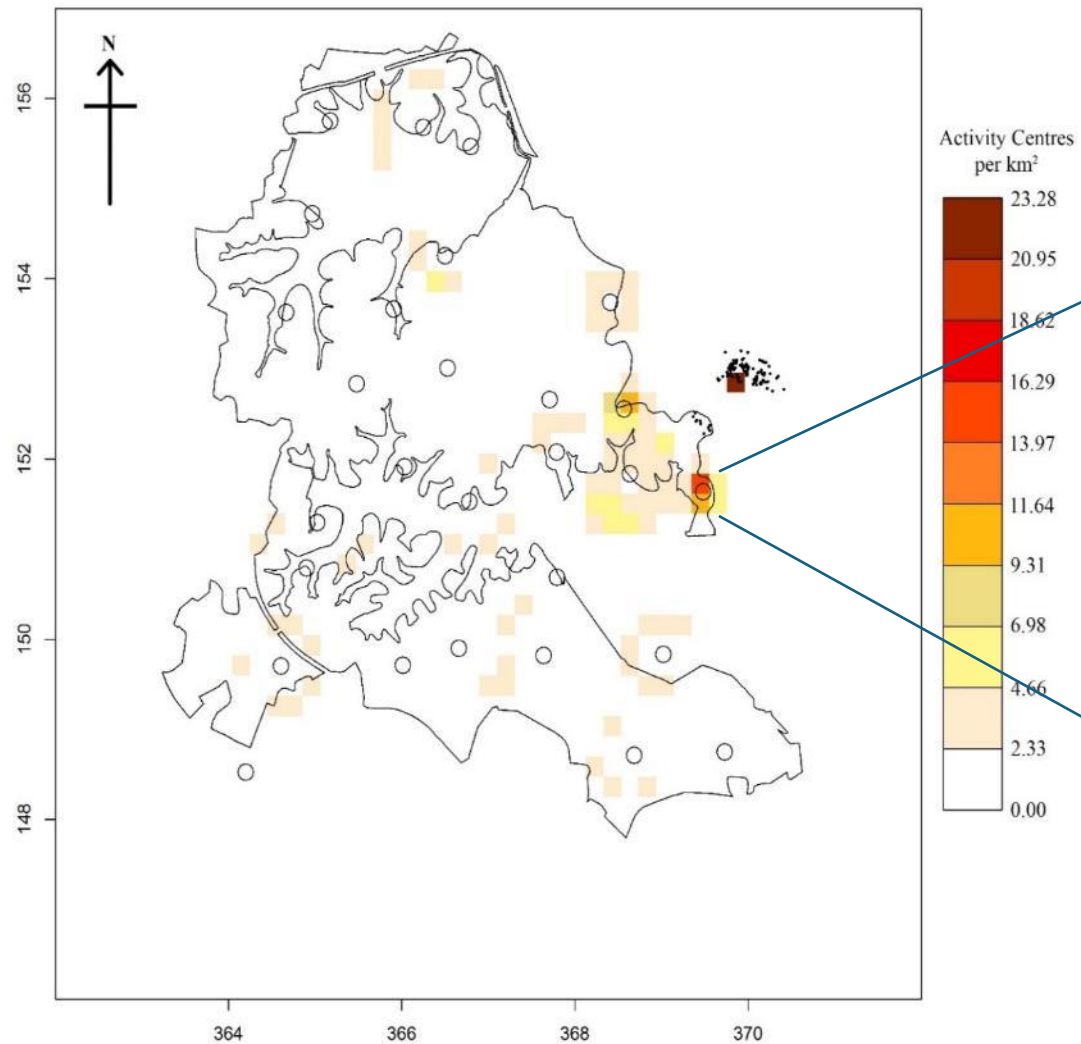


PUB and NParks trapped a 1.7m-long crocodile at Lower Seletar Reservoir on Feb 23, 2019. PHOTO: PUB/FACEBOOK

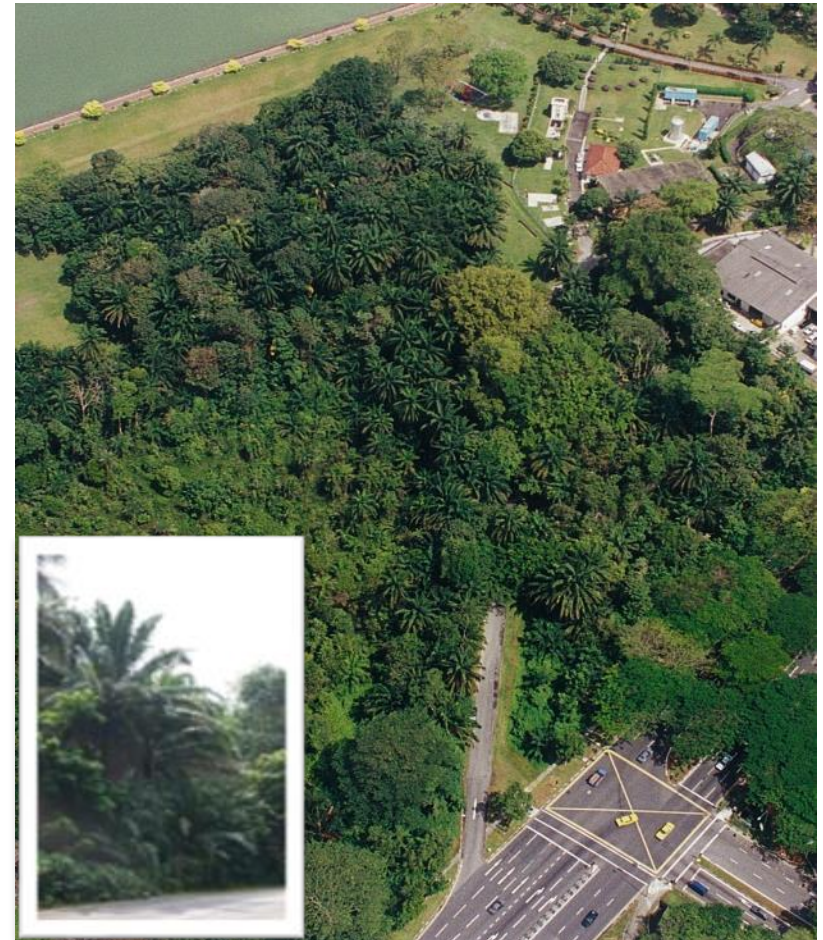
A 1.7m-long crocodile was trapped in the early hours of Feb 23 by national water agency PUB and NParks.

Population Management

Case study: Wild boars



Hotspot for wild boars coincide with ex-oil palm plantations at Lower Peirce



Koh et al. (2018)

Wildlife Population & Distribution Mapping



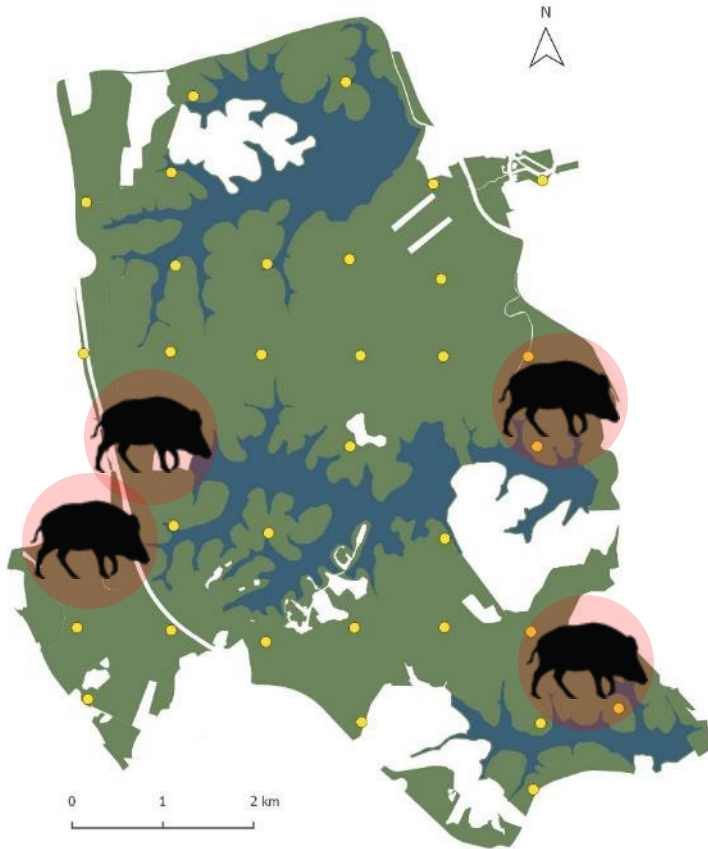
Wild boar



Sambar deer



Lesser mouse-deer



- Areas with higher detections of each species are highlighted



Original Research Article

Persistence of a locally endangered mouse-deer amidst the re-emergence of two larger ungulates in small urban rainforest fragments

Max D.Y. Khoo^{*}, Bryan T.M. Lim, Malcolm C.K. Soh, Rebecca H.Y. Loy, H.K. Lua, Benjamin P.Y.-H. Lee, Adrian H.B. Loo, Kenneth B.H. Er

National Parks Board, 1 Cluny Road, Singapore Botanic Gardens, 259569, Singapore

ARTICLE INFO

Keywords:

Conservation monitoring
Wildlife management
Urban ecology
Random encounter
Model
Two-species occupancy
Competition

ABSTRACT

Urbanisation has threatened native forests globally with remaining forests often existing as small isolated patches in a harsh landscape matrix. Devoid of large predators, forest patches may continue to support ungulate populations, but their conservation is challenging as unchecked population growth can bring about ecosystem imbalances and threaten other native species. Three ungulate species exist in small rainforest fragments of urbanised Singapore: the locally endangered lesser mouse-deer (*Tragulus kanchil*), and the re-emergent wild boar (*Sus scrofa*) and sambar deer (*Rusa unicolor*). Using an array of camera traps, we estimated the densities of the lesser mouse-deer, wild boar and sambar deer in rainforest patches, and determined the influence of habitat and landscape factors on their occupancies. We also studied the interactions of the wild boar and sambar deer with the lesser mouse-deer spatially and temporally. Estimates of the lesser mouse-deer density showed that the population has recovered since the 1990s. Its density was comparable to the wild boar, but higher than sambar deer. Detections of lesser mouse-deer were significantly higher in primary forests compared to old secondary forests, and increasing distance away from the nearest road or public trail was a strong predictor of its occupancy. There was no evidence that the wild boar and sambar deer competitively excluded the lesser mouse-deer, rather, co-occupancy was observed. We demonstrated that temporal niche partitioning between the lesser mouse-deer and the two other ungulates likely explained their co-existence. Notwithstanding this, there are indications of understorey alteration due to the wild boar occurrences. Thus, continued monitoring of ungulate populations and conservation efforts focused on maintaining the health of the primary and secondary rainforest patches, and reducing disturbance, are critical to sustain future populations of lesser mouse-deer. Our study has shed new light on the habitat use and species interactions of tropical ungulates in urban predator-free ecosystems and highlight the importance in gaining a deeper understanding of their dynamics to more effectively guide conservation and management decisions.

11

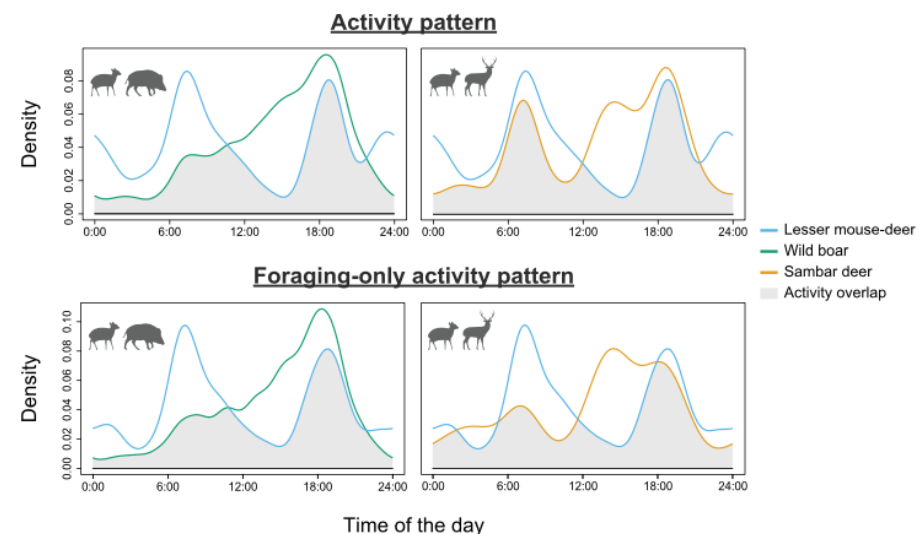


Fig. 5. Overall activity patterns and foraging-only activity patterns of the species pairs. The coefficient of overlap is represented by the grey area under the curves. Daylight hours are roughly between 0700 and 1900 h during our period of study.

The two larger ungulate species did not competitively exclude the lesser mouse-deer, which is likely due to temporal niche partitioning or because the wild boar and sambar deer densities are still relatively low.



“Being a biophilic city is also about how communities become part of the ecosystem and stewards of nature”



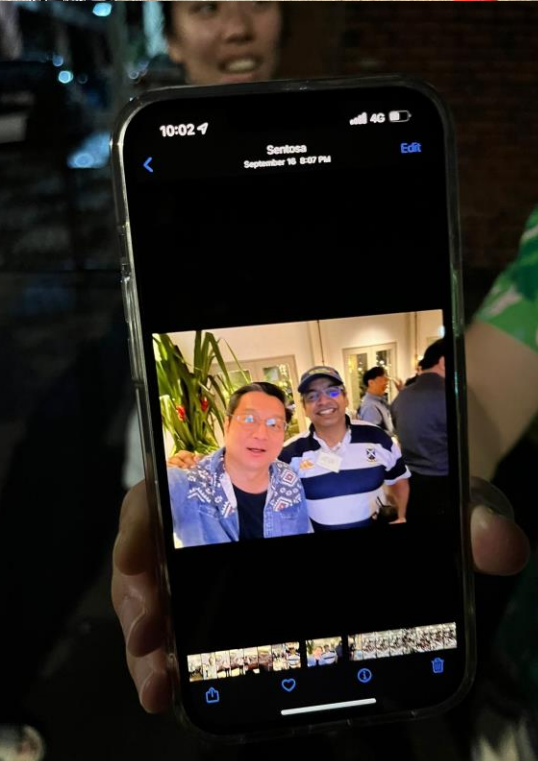
2022-10-05 8:34:43 PM M 1/10



2022-10-05 8:34:44 PM M 2/10



2022-10-05 8:34:44 PM M 2/10



Singapore

Relocation of otters from Seletar was 'last resort' given the presence of pups and location of burrow: Experts

Setting up barriers to prevent the otters from accessing private estates and ponds is still a critical first measure, said experts.



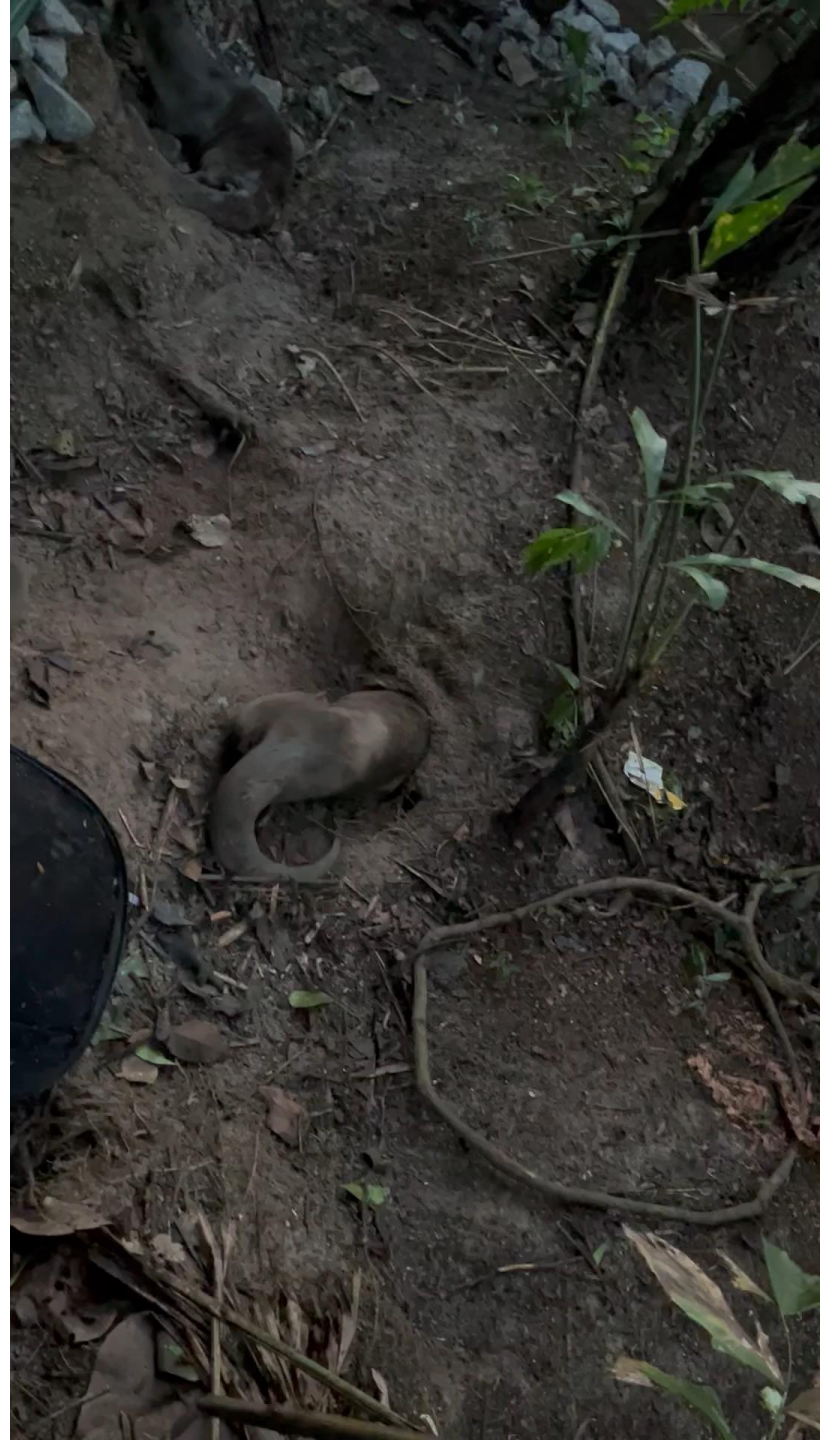
Otters at Seletar estate in Singapore. (Photo: Otter Working Group/Marjorie Chong)



Koh Wan Ting

18 Oct 2022 05:43PM
(Updated: 18 Oct 2022 06:05PM)

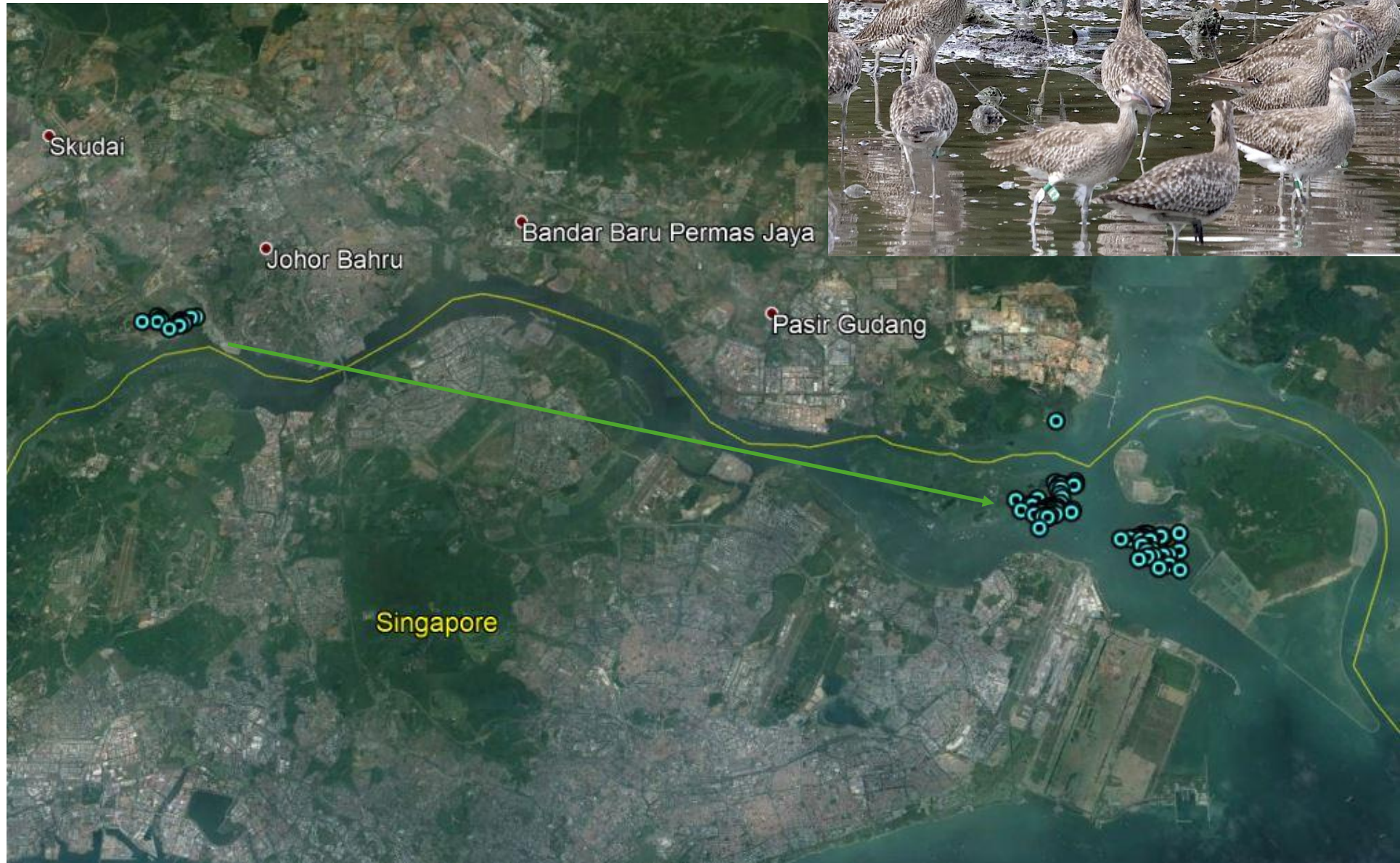




4. The bird that needed a thigh massage

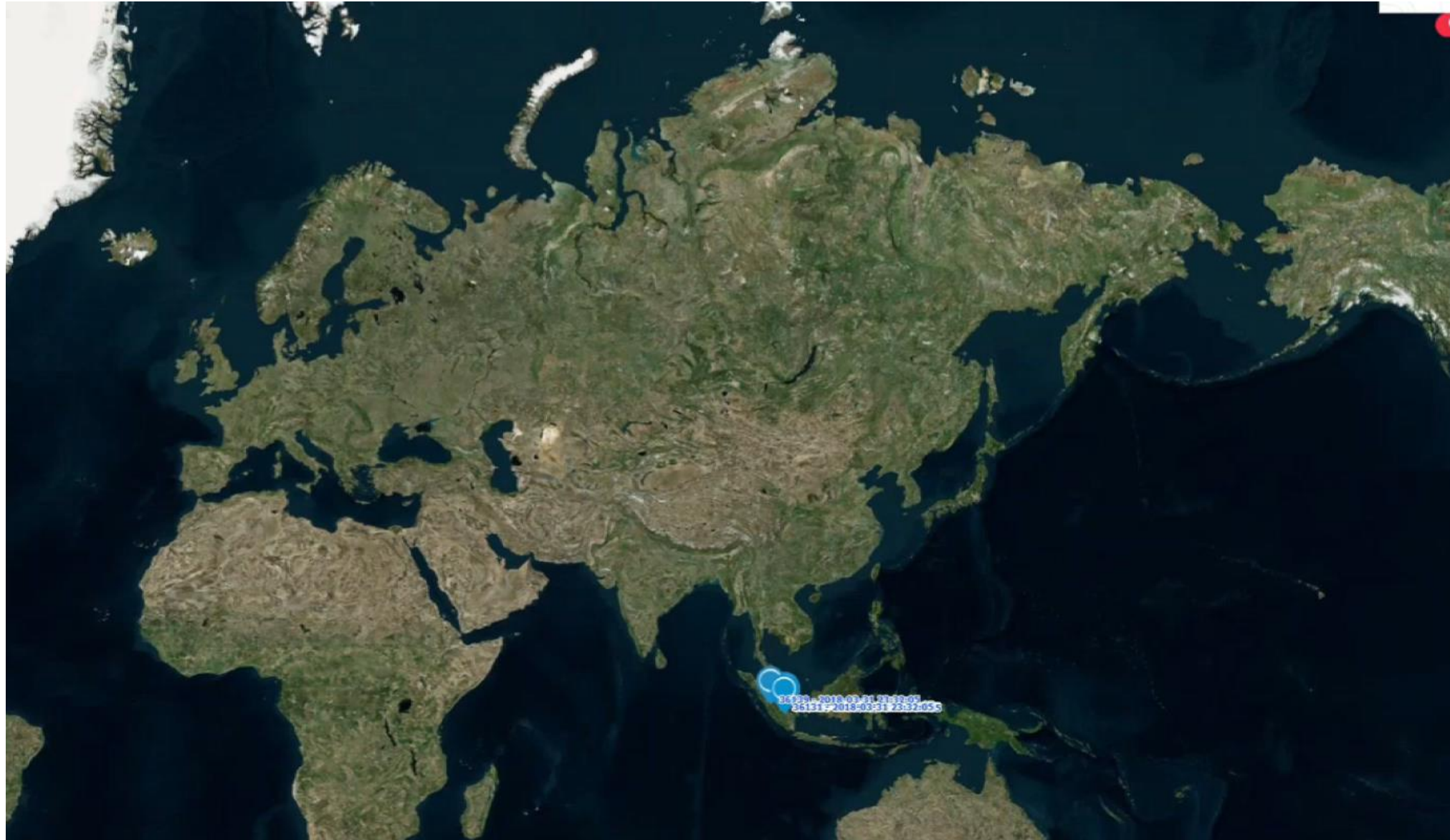


Whimbrel E6



Satellite tracking of migratory shorebirds

-  Common Greenshank B7
-  Common Redshank WU
-  Common Redshank A00
-  Common Greenshank B6
-  Pacific Golden Plover F3
-  Common Redshank EP
-  Grey Plover B3
-  Whimbrel E9
-  Whimbrel E6
-  Whimbrel E7
-  Whimbrel E8





Arctic Council



Working Group

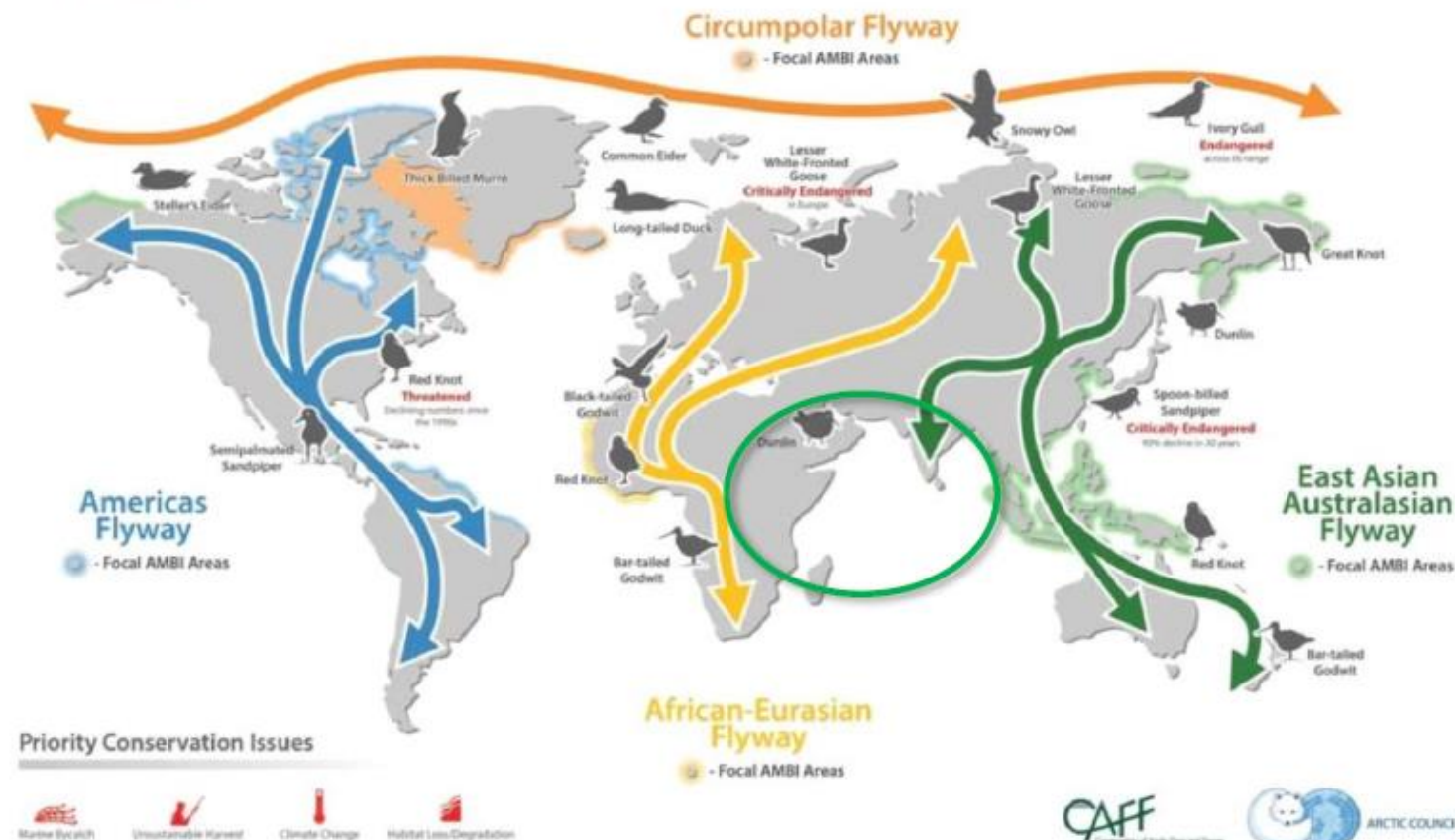


Arctic Migratory Birds Initiative

Strategies

The Arctic Migratory Birds Initiative (AMBI)

The Arctic Migratory Birds Initiative (AMBI) is a project under the Conservation of Arctic Flora and Fauna (CAFF) designed to improve the status and secure the long-term sustainability of declining Arctic breeding migratory bird populations. AMBI works with many partner countries and organizations to support and initiate actions that target **priority species** and **conservation issues** across a series of **focal geographies**. For more information please visit: www.caff.is/ambi





12th Ministerial Meeting of the Arctic Council

20 May 2021 | Observer statement

SINGAPORE

Singapore as Observers in the Arctic Council

SINGAPORE'S STATEMENT FOR THE ARCTIC COUNCIL MINISTERIAL MEETING, MAY 2021

programmes for Permanent Participants. The Council can also count on Singapore's active participation in the Conservation of Arctic Flora and Fauna Working Group on the Arctic Migratory Birds Initiative (AMBI) project. With a growing number of Singaporean youths taking an interest in the Arctic, we will also explore possible synergies to strengthen the Arctic Council's youth engagement.

Minister of State in the Prime Minister's Office and Ministry of Manpower, Mr Sam Tan, opened the workshop at [Sungei Buloh Nature Reserve](#) [EAAF073], a key stopover for Arctic migratory birds.

The Arctic Migratory Birds Initiative is a project under the [Arctic Council's](#) Conservation of Arctic Flora and Fauna working group. Singapore has been a permanent observer on the Arctic Council since 2013.

Participants will discuss priorities for intertidal habitat preservation in Southeast Asia, Yellow Sea conservation, demonstration projects and illegal bird hunting.

5. The scale of pangolin trafficking



Record haul of pangolin scales worth \$52.3m seized in Singapore



World record haul of pangolin scales worth \$52 million seized from container at Pasir Panjang



These pangolin scales from Africa, squeezed into 230 sacks, were found hidden behind packets of frozen beef inside a container at the Pasir Panjang Export Inspection Station on Wednesday. The 12.9 tonnes of scales worth \$52.3 million may have come from 17,000 pangolins, and is the biggest single haul in the world in recent years. The scales are in demand in Asia for use in traditional Chinese medicine despite there being no proven medical benefit from their use. ST PHOTO: ONG HEE JIN SEE: SEE HOME 34

3 April 2019 – Record Pangolin Scale seizure

- 12.9 tonnes of pangolin scales worth \$52.3 million
- 230 bags among packets of frozen beef,
- 177kg of cut up and carved elephant ivory worth \$120,000
- Joint ops with Singapore Customs
- Word record haul
- Nigeria to Vietnam
- Scales from 17,000 pangolins



Hidden among packets of frozen beef



230 bags of scales from 17,000 pangolins



177kg of elephant ivory





8 April 2019 – second seizure in just 5 days

- 12.7 tonnes of pangolin scales, packed in 474 bags, worth S\$51.6 million
- 40-footer container that was on its way from Nigeria to Vietnam
- Declared as Cassia seeds
- 21,000 pangolins
- *Smutsia gigantea* (Giant Ground Pangolin) – 1007
- *Phataginus tricuspis* (White-Bellied Tree Pangolin) - 20408



THE STRAITS TIMES

SINGAPORE

Second 12-tonne haul of pangolin scales seized in less than a week



Vanessa Liu

UPDATED APR 11, 2019, 06:14 AM



SINGAPORE - Less than a week after a record 12.9 tonnes of pangolin scales were seized from a container by the authorities here, another 12.7 tonnes have been uncovered in a joint operation by the National Parks Board (NParks), Singapore Customs and the Immigration and Checkpoints Authority (ICA).

The 40ft (12m) container that was on its way by ship from Nigeria to Vietnam was declared to contain "cassia seeds" but was found on Monday (April 8) to be holding the animal parts, estimated to be worth about \$51.6 million, in 474 bags at the Pasir Panjang Scanning Station.



21 July 2019

- 11.9 tonnes of pangolin scales, packed in 237 bags and worth about S\$48.6 million
- 8.8 tonnes of raw elephant ivory, packed in 137 bags and worth about S\$17.6 million
- Democratic Republic of the Congo to Vietnam
- shipment declared as containing "timber"
- Estimated 2000 pangolins and 300 elephants

THE STRAITS TIMES

SINGAPORE



Vanessa Liu

UPDATED JUL 27, 2019, 10:14 PM ▾



SINGAPORE - A record 8.8 tonnes of elephant ivory was seized from a shipment on Sunday night (July 21) in what is believed to be one of the largest seizures that the world has seen in recent years.

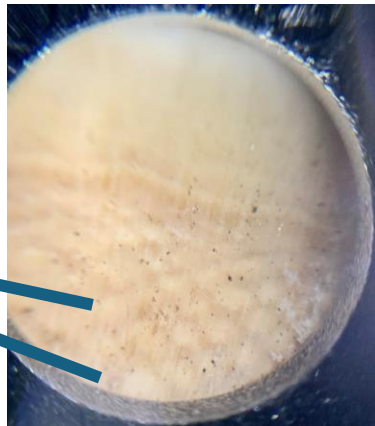
The ivory is estimated to have come from nearly 300 African elephants. Just three months earlier in April, Singapore seized 177kg of cut-up and carved elephant ivory.

Alongside the record ivory haul, 11.9 tonnes of pangolin scales were also confiscated in the operation. It was the third shipment to be intercepted this year, said the National Parks Board (NParks), Singapore Customs and the Immigration and Checkpoints Authority (ICA) in a joint statement.

Singapore

Singapore seizes record haul of ivory alongside pangolin scales in S\$66m shipment





As an international trading hub, Singapore is committed to tackle illegal wildlife trade



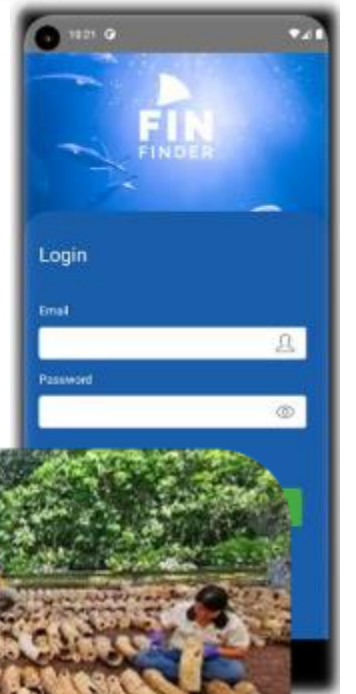
Advanced investigation – *Centre for Wildlife Forensics*



Aims to become an international
centre of excellence in wildlife
forensic research and collaboration.

Technology streams

- DNA analysis (Identification and Geographic analysis of illegal trade and poaching)
- Vision-based/AI-assisted identification (Xylotron, FIN Finder)
- Spectrographic and Chemical Analysis (DART)



Documenting and sampling of large-scale ivory seizures

- In collaboration with University of Washington.
- Documenting ivory tusks.
- Matching of ivory pairs.
- Sampling for DNA extraction.



Elephant genotypes reveal the size and connectivity of transnational ivory traffickers

Samuel K. Wasser¹✉, Charles J. Wolock², Mary K. Kuhner¹, John E. Brown III³, Chris Morris⁴, Ryan J. Horwitz⁵, Anna Wong⁶, Charlene J. Fernandez⁶, Moses Y. Otiende⁷, Yves Hoareau¹, Zofia A. Kaliszewska¹, Eunjin Jeon¹, Kin-Lan Han¹ and Bruce S. Weir²

Transnational ivory traffickers continue to smuggle large shipments of elephant ivory out of Africa, yet prosecutions and convictions remain few. We identify trafficking networks on the basis of genetic matching of tusks from the same individual or close relatives in separate shipments. Analyses are drawn from 4,320 savannah (*Loxodonta africana*) and forest (*L. cyclotis*) elephant tusks, sampled from 49 large ivory seizures totalling 111t, shipped out of Africa between 2002 and 2019. Network analyses reveal a repeating pattern wherein tusks from the same individual or close relatives are found in separate seizures that were containerized in, and transited through, common African ports. Results suggest that individual traffickers are exporting dozens of shipments, with considerable connectivity between traffickers operating in different ports. These tools provide a framework to combine evidence from multiple investigations, strengthen prosecutions and support indictment and prosecution of transnational ivory traffickers for the totality of their crimes.

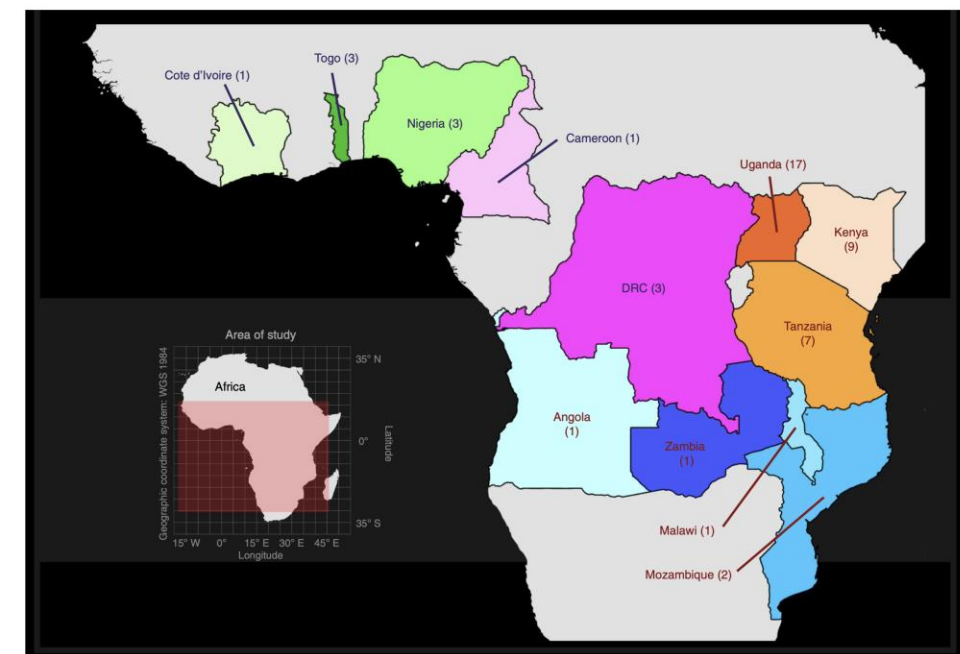


Fig. 1 | Geographical span and colour code scheme for ports of containerization used for all consequent figures. Number of seizures per port is shown in parentheses. Map displayed using the World Geodetic System (WGS) 1984 geographic co-ordinate system. Country boundary data are from Esri, Garmin International, US Central Intelligence Agency and National Geographic Society. World Countries (Generalized) (feature class). ArcGIS Living Atlas of the World. April 2021.

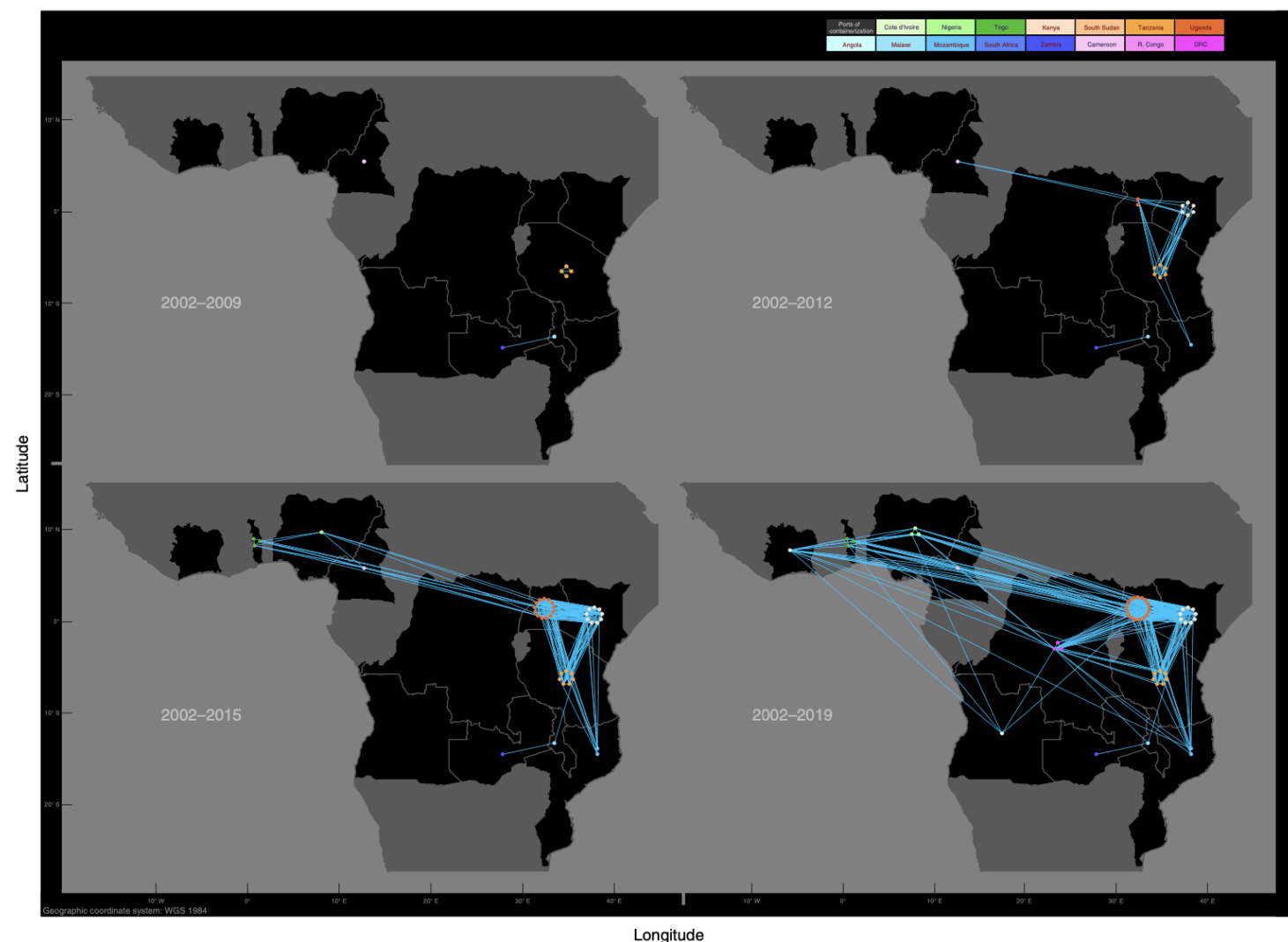
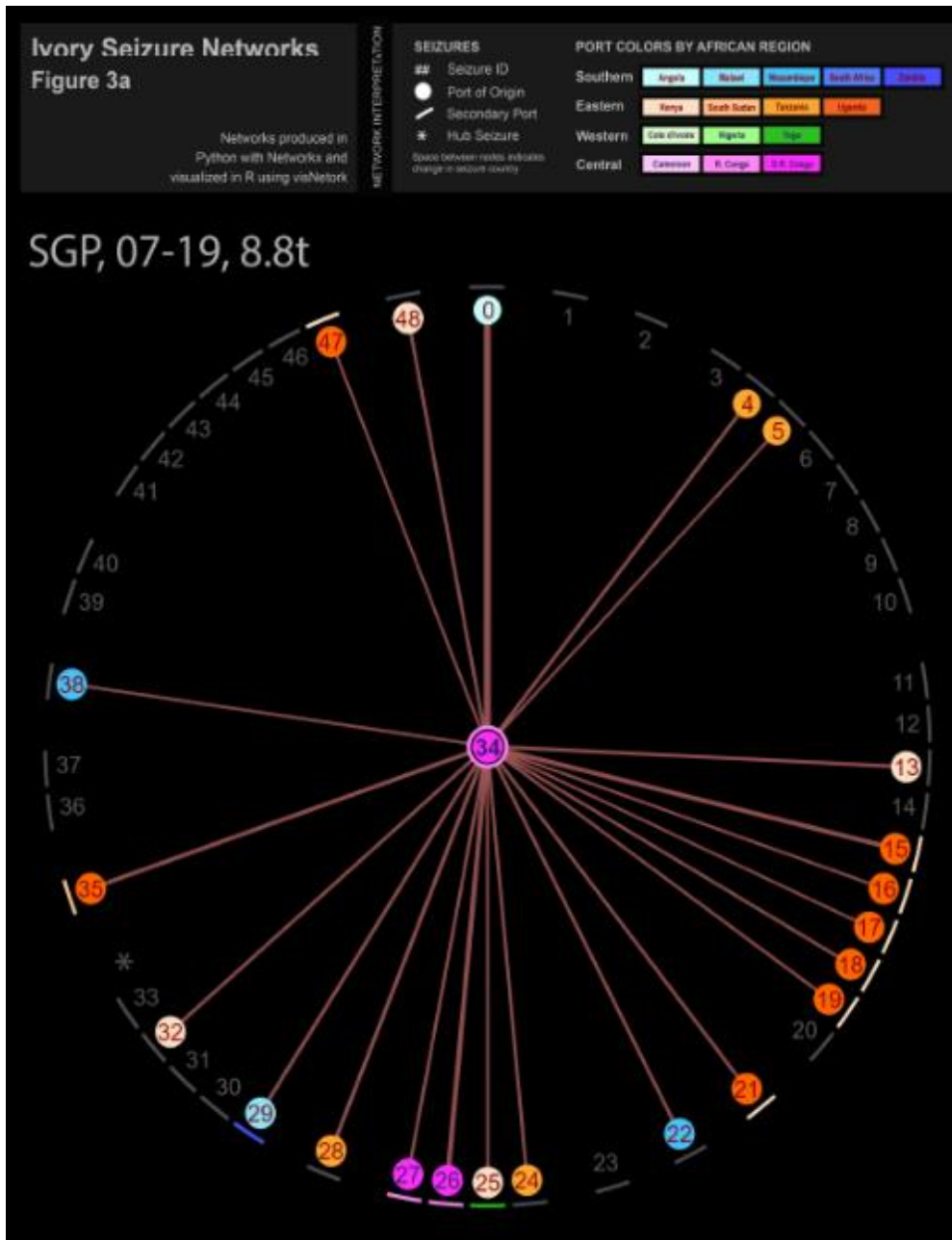


Fig. 2 | Temporal progression in the major ports of containerization and their connectivity, based on genetic matches and shared physical evidence between seizures. Each seizure is represented by a solid point with the port of containerization indicated by the colour of that point as per the colour scheme in Fig. 1. A blue line connecting any two seizures indicates one or more genetic matches and/or shared physical evidence between them. Points are centred around country centroids and arranged in circles for ease of visualization, with circle area being proportional to the number of points in a given country. Early seizures of savannah elephant ivory were containerized in Tanzania, Zambia and Malawi but shifted to Kenya (2010–2012) and then Uganda (2013–2015). Containerization next shifted to western Central Africa (DRC and Angola) (2016–2019). Connectivity within and between all of these shipments grew with the number of new seizures. Similarly, ports of containerization of forest elephant ivory shifted from Togo (2013–2014) to Nigeria (2016–2019) along with connections to each other but also to multiple seizures containerized in East and western Central Africa. Country boundary data are from the same source as Fig. 1. Map displayed using the WGS 1984 co-ordinate system. R. Congo, Republic of the Congo.



Case example - Singapore seizure in July 2019

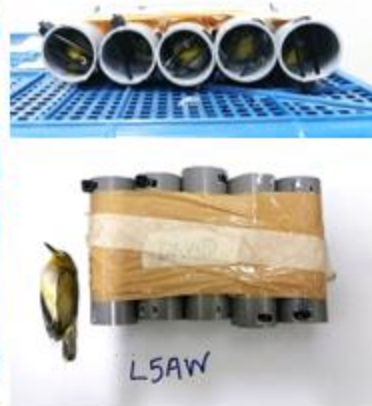
- Has tusks that are genetically related to seizures from various port of origins
- Strongest link to seizure #0 (Angola), #26 (DRC) and #29 (Angola)
- Tusks could come from a herd from the KAZA region
- Could be linked to a common syndicate collecting tusks from poachers
- Focus enforcement on hotspot
- Convict syndicate for larger crime

Our own backyard - Bird Smuggling

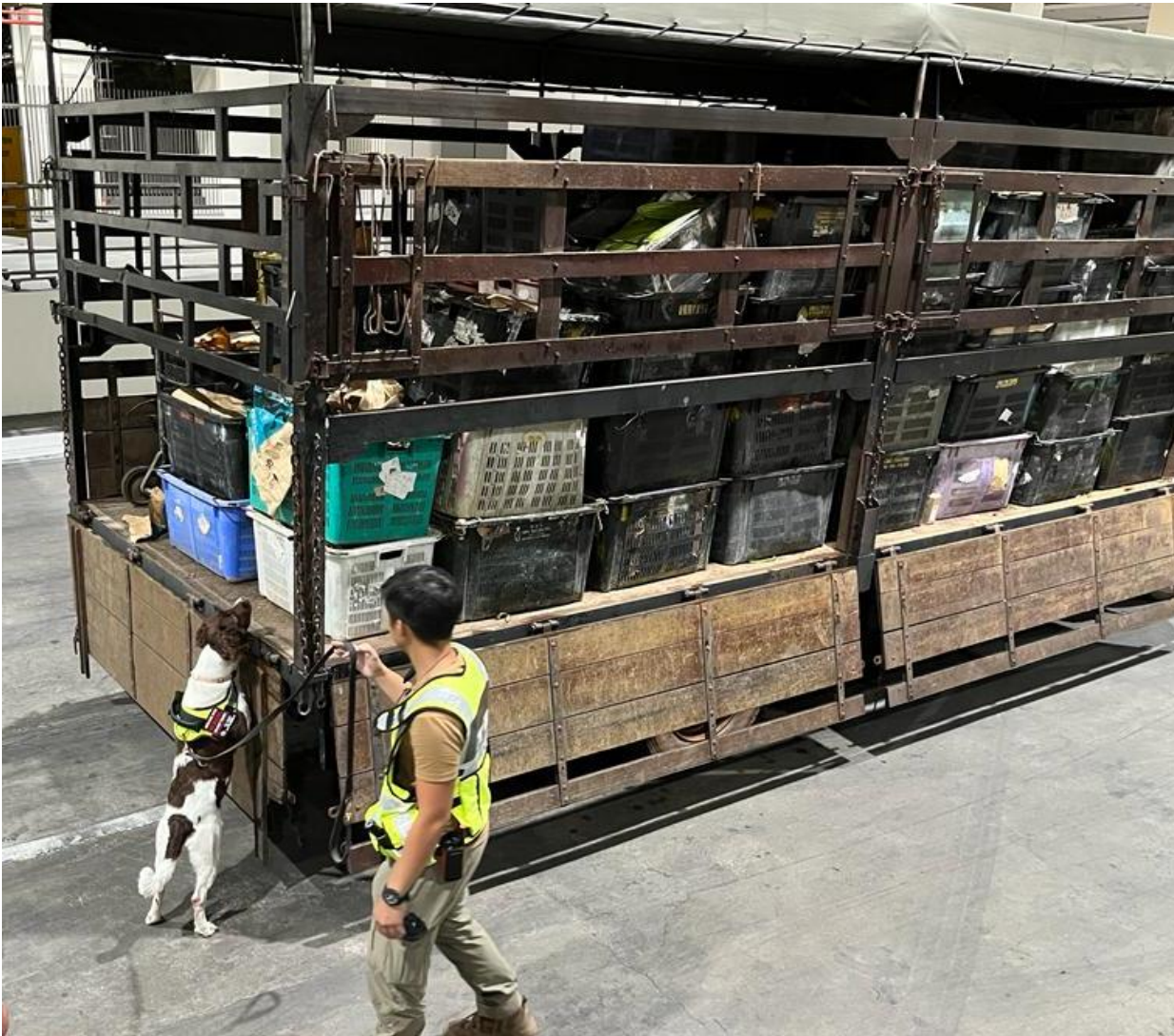


Record 815 birds found in bus as 'chirping sounds' foil smuggling attempt at Woodlands Checkpoint

Date	August 2019
Port of entry	Woodlands Checkpoint
Offender	A 35-year old Malaysian male Tour bus driver entering Singapore from Malaysia
Modus Operandi	Modified hidden compartment above the left and right rear tyres concealing 15 trays and 5 tubes. Birds confined in small spaces in high density.
Birds smuggled	626 × Munia 141 × Swinhoe's White-Eye 38 × White-Rumped Shama 10 × Oriental Magpie Robin 815 birds in total, 166 (20%) dead on arrival



Enforcement and risk assessment



Formation of a K-9 unit dedicate to sniff out IWT at our borders

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Singapore

Pet shop owner linked to Singapore's largest bird-smuggling operation gets jail

Leow Seng Lee, who has been in the pet shop business for more than 40 years, worked with a middleman who hired a Malaysian truck driver to smuggle birds across the border.

Lydia Lam
@LydiaLamCNA
08 Apr 2024 04:23PM

ICA officers detected eight plastic containers containing 337 birds in a trailer truck (left). (Photos: Court house with CNA file)

FAST

Stronger enforcement actions against online sellers

- Annual island-wide operations
- Targeted at online exotic pets sellers on platforms such as telegram
- Internet and Social Media Marketplaces

THE STRAITS TIMES

SINGAPORE

▲ LC

RSAF regular who sold pregnant pangolin gets probation



Damien Tan Guan Rong is to remain indoors from 11pm to 6am every day. ST PHOTO: KELVIN CHIU

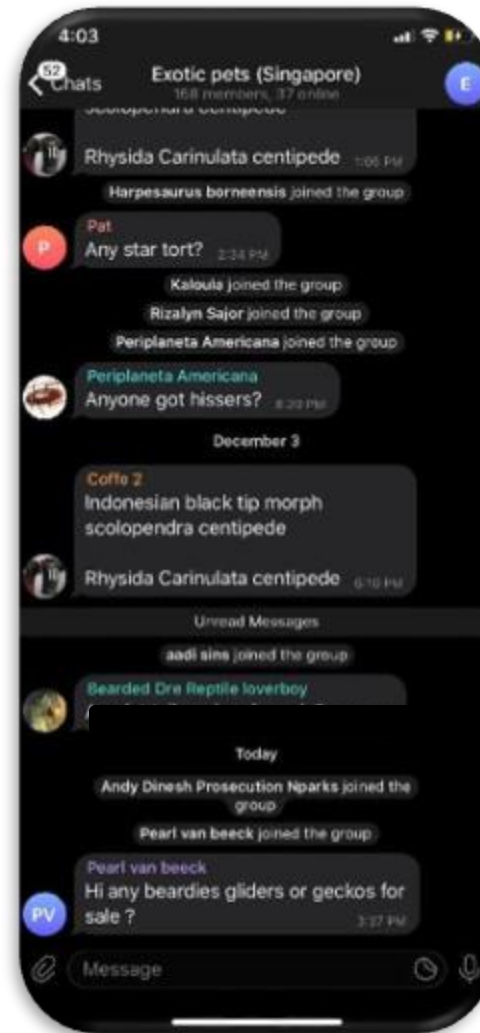
Christine Tan

UPDATED MAR 01, 2024, 03:49 PM

SINGAPORE - A Republic of Singapore Air Force regular was sentenced to a year's probation on Feb 29 for selling a pangolin he found at a park connector.

Damien Tan Guan Rong, 21, who holds the rank of Military Expert 1 and is employed as a technician, is to remain indoors from 11pm to 6am every day.

If the curfew period is required to be varied for official work duties, it must be with prior approval from his work supervisors.





H

Baby hedgehog 🦿 , pm if interested! 4:41 PM

00:03



Y

Dark red bearded dragon \$350 interested pm
Cash and carry 4:46 PM

00:20



00:14



5:09 PM



clearing 10:52 AM



T

\$130. Male

11:09 AM

00:07



NLB Albino Patternless knob tail

👉 Male

👉 1 year + old

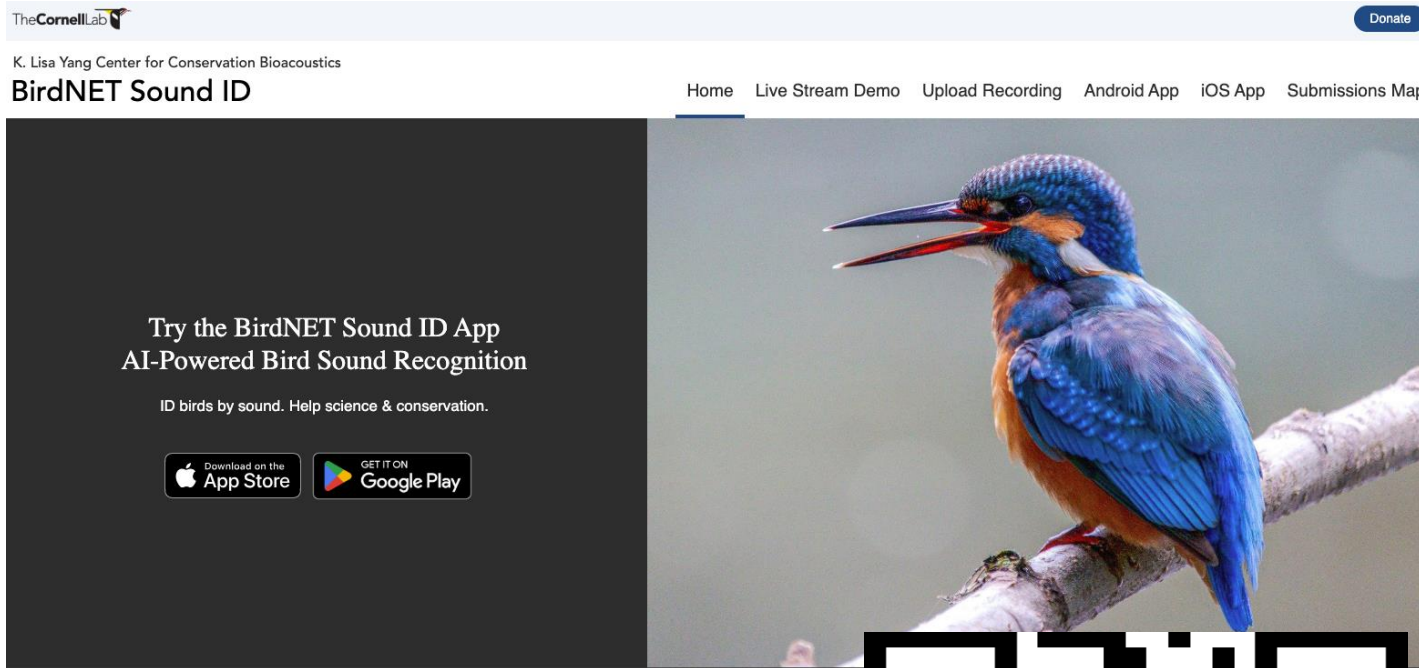
✉ For more details

11:32 AM

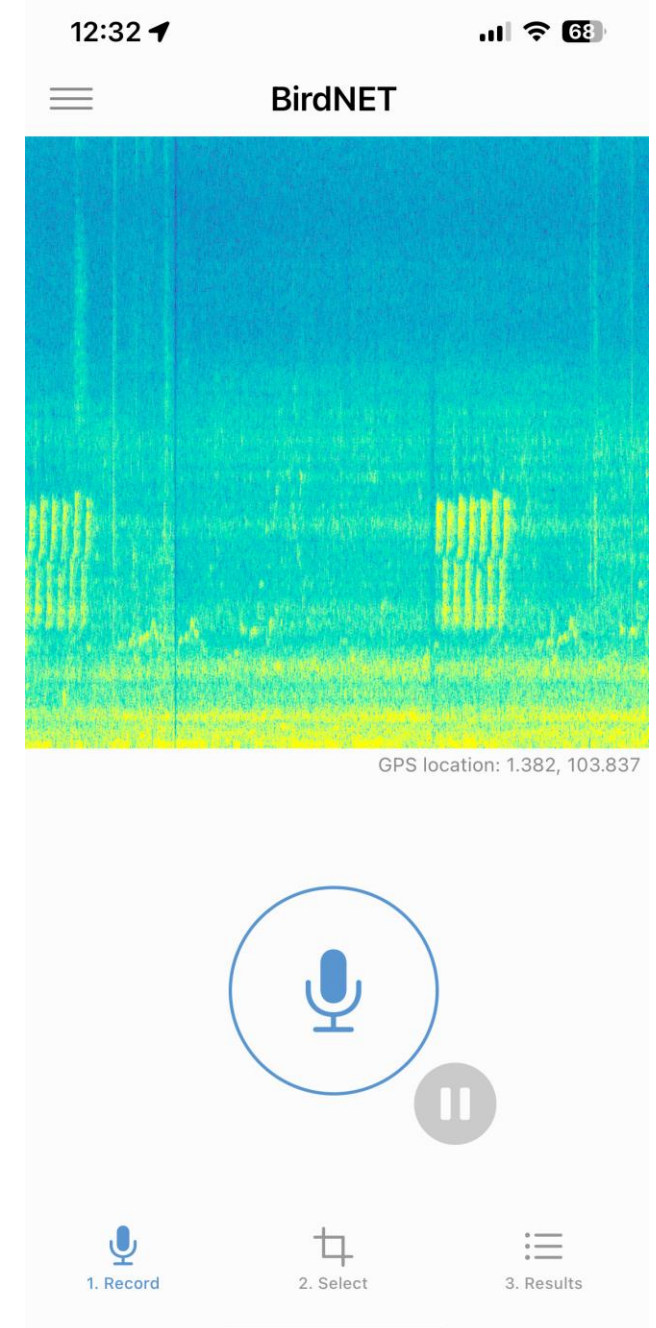




6. Songs of the forest



<https://birdnet.cornell.edu/>



Please download on your phone







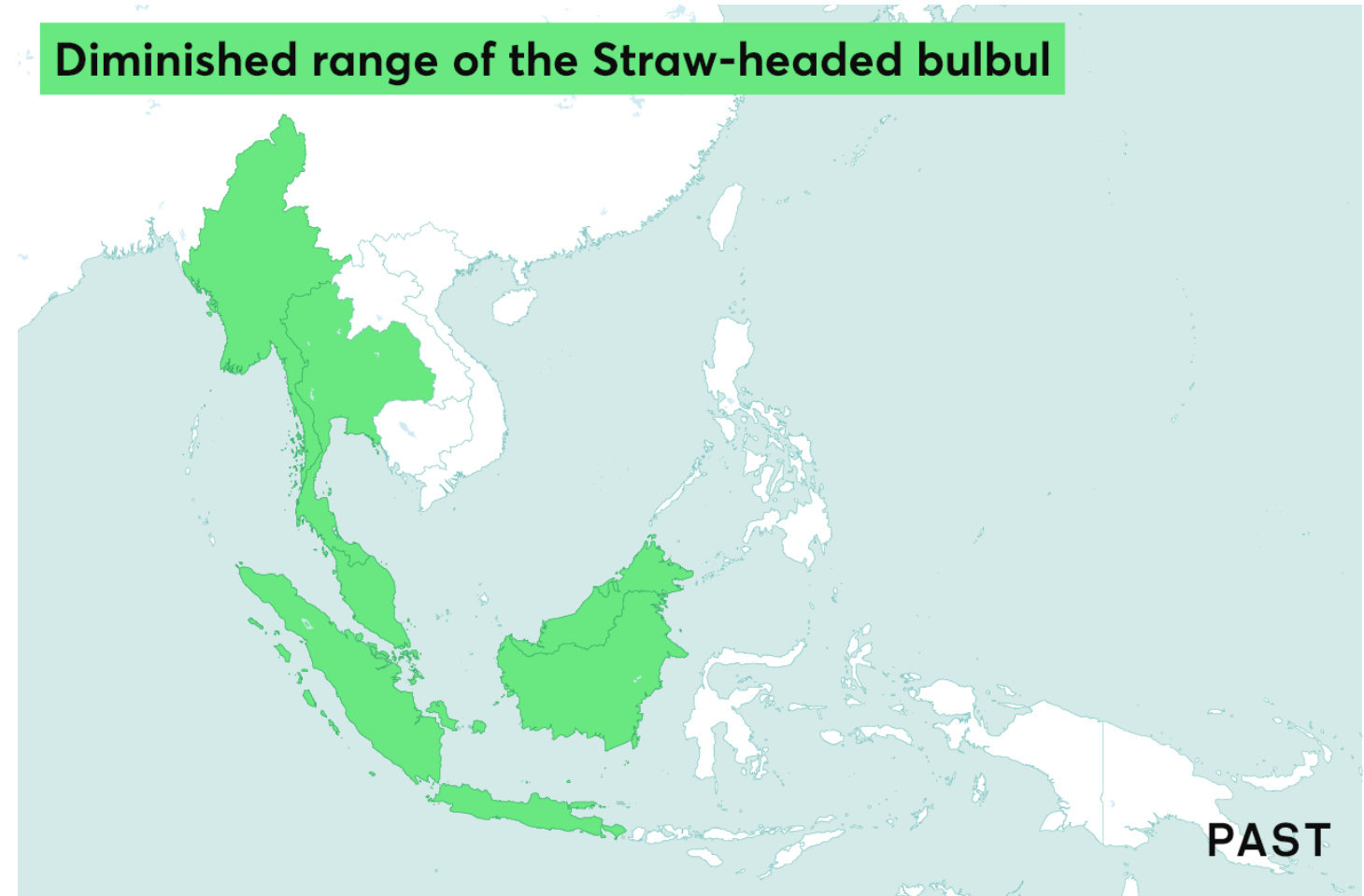
Straw-headed Bulbul
(*Pycnonotus zeylanicus*)



Of the 337 birds smuggled in, 138 of them died, including the endangered white-rumped shama – a popular songbird in South-east Asia that has become increasingly rare in the region due to its popularity in the pet trade. PHOTO: FRANCIS YAP

Serious threats in many parts of the region

- SHB is listed in CITES Appendix II, but the **population is still declining due to excessive trapping and illegal trade.**
- Believed to be extirpated from Thailand, Myanmar, and Java and Sumatra in Indonesia







The side-events

SINGAPORE'S CENTRE FOR WILDLIFE FORENSICS



Tracing Second Pangolin Scales to the Source



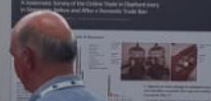
Singapore's Centre for Wildlife Forensics



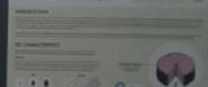
Fauna Identification and Analysis



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Straw-headed bulbul



Singapore, Malaysia, and the United States of America propose to uplist the Straw-headed Bulbul (*Pycnonotus zeylanicus*) to CITES Appendix I

The Straw-headed Bulbul is critically endangered globally and heavily trapped across Southeast Asia due to high demand in the bird trade for its distinctive and melodious singing

White-rumped Shama

Malaysia & Singapore propose to uplist the White-rumped Shama (*Copsychus malabaricus*) to CITES Appendix II

The White-rumped Shama is becoming increasingly rare throughout the region because of its popularity in the pet trade

Over 32,000 birds were illegally trafficked from 2009-2022

Why Appendix II listing is needed?

Improve monitoring of trade and to provide information for effective regulation





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Thank you